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Cloud Computing Lab

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Section: 5-B

LAB NO 9

Codespaces + AWS: GH CLI (Codespaces), AWS CLI, EC2, IAM, Security Groups, Filters & Queries

Lab Tasks:

Task 1 — GitHub CLI, Codespace setup and authentication

1. Local desktop) Install GitHub CLI (Windows example via winget):

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\saira> winget install --id GitHub.cli
The 'msstore' source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function properly (ex. "US").

Do you agree to all the source agreements terms?
[Y] Yes [N] No: yes
Found GitHub CLI [GitHub.cli] Version 2.83.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/cli/cli/releases/download/v2.83.1/gh_2.83.1_windows_amd64.msi
17.6 MB / 17.6 MB
Successfully verified installer hash
Starting package install...
Successfully installed
PS C:\Users\saira> |
```

2. (Local) Authenticate GH CLI for Codespaces:

```
PS C:\Users\saira> gh auth login -s codespace
? Where do you use GitHub? GitHub.com
? What is your preferred protocol for Git operations on this host? HTTPS
? Authenticate Git with your GitHub credentials? Yes
? How would you like to authenticate GitHub CLI? Paste an authentication token
Tip: you can generate a Personal Access Token here https://github.com/settings/tokens
The minimum required scopes are 'repo', 'read:org', 'workflow'.
? Paste your authentication token: - gh config set -h github.com git_protocol https
✓ Configured git protocol
✓ Logged in as 23-22411-057-sudo
PS C:\Users\saira> |
```

3. (Local) List available Codespaces (optional verification):

```
Token scopes: admin:org, codespace, repo
PS C:\Users\saira> gh codespace list
NAME              DISPLAY NAME      REPOSITORY          BRANCH  STATE    CREATED AT
fluffy-pancake-69x9r4r4j... fluffy pancake     23-22411-057-sudo/... main*   Shutdown about 1 month ago
probable-space-giggle-97... probable space giggle 23-22411-057-sudo/... main    Shutdown about 29 days ago
PS C:\Users\saira>
```

```
You were already logged in to this account
PS C:\Users\saira> gh auth status
github.com
✓ Logged in to github.com account 23-22411-057-sudo (keyring)
- Active account: true
- Git operations protocol: https
- Token: ghp_*****
- Token scopes: 'admin:org', 'codespace', 'repo'
```

4. (Local) Create or connect to a Codespace. To create a new codespace from the current repo:

```
PS C:\Users\saira> gh codespace create --repo 23-22411-057-sudo/exam_prep --branch main --machine basicLinux32
gb
✓ Codespaces usage for this repository is paid for by 23-22411-057-sudo
studious-fishstick-r454rprpjx94f6p7
```

Or to open an existing codespace:

```
PS C:\Users\saira> gh codespace ssh -c studious-fishstick-r454rprpjx94f6p7
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1030-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

Task 2 — Install AWS CLI inside the Codespace and configure it

1. Download and install AWS CLI:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left   Speed
100 60.0M  100 60.0M    0     0  129M      0 --:--:-- --:--:-- --:--:-- 129M
```

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ unzip awscliv2.zip
Archive: awscliv2.zip
  creating: aws/
  creating: aws/dist/
  inflating: aws/README.md
  inflating: aws/install
  inflating: aws/THIRD_PARTY_LICENSES
  creating: aws/dist/awscli/
  creating: aws/dist/dateutil/
  creating: aws/dist/docutils/
  creating: aws/dist/lib-dynload/
  creating: aws/dist/prompt_toolkit-3.0.51.dist-info/
  creating: aws/dist/wheel-0.45.1.dist-info/
  inflating: aws/dist/aws
  inflating: aws/dist/aws_completer
  inflating: aws/dist/libpython3.13.so.1.0
  inflating: aws/dist/_awscli.abi3.so
  inflating: aws/dist/_ruamel_yaml.cpython-313-x86_64-linux-gnu.so
  inflating: aws/dist/libz.so.1
  inflating: aws/dist/liblzma.so.5
  inflating: aws/dist/libbz2.so.1
  inflating: aws/dist/libffi.so.6
  inflating: aws/dist/libuuid.so.1
  inflating: aws/dist/libreadline.so.6
  inflating: aws/dist/libtinfo.so.5
  inflating: aws/dist/libsqlite3.so.0
```

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

2. Verify installation:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws --version
aws-cli/2.32.10 Python/3.13.9 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

3. Configure the AWS CLI (you will provide Access Key ID and Secret Access Key for a user with permissions, or use root/IAM user you prepared for the lab):

Step 1
Access key best practices & alternatives
Step 2 - optional
Set description tag
Step 3
Retrieve access keys

Set description tag - optional [Info](#)

The description for this access key will be attached to this user as a tag and shown alongside the access key.

Description tag value

Describe the purpose of this access key and where it will be used. A good description will help you rotate this access key confidently later.

Maximum 256 characters. Allowed characters are letters, numbers, spaces representable in UTF-8, and: _ . : / = + - @

[Cancel](#)
[Previous](#)
[Create access key](#)

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws configure
AWS Access Key ID [None]: AKIAVHLXGUTLDQSQTAH3
AWS Secret Access Key [None]: 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
Default region name [None]: me-central-1
Default output format [None]: json
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

4. Verify credentials/config files:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat ~/.aws/credentials
[default]
aws_access_key_id = AKIAVHLXGUTLDQSQTAH3
aws_secret_access_key = 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
@23-22411-057-sudo → /workspaces/exam_prep (main) $
@23-22411-057-sudo → /workspaces/exam_prep (main) $ cat ~/.aws/config
[default]
region = me-central-1
output = json
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

5. Verify connectivity (you should see a JSON result showing your caller identity):

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws sts get-caller-identity
{
  "UserId": "AIDAVHLXGUTLK25C57B4F",
  "Account": "359416636630",
  "Arn": "arn:aws:iam::359416636630:user/Admin"
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

Task 3 — Create security group and add ingress rules using Codespace IP

1. Create a security group (substitute your VPC id):

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 create-security-group \
--group-name 'MySecurityGroup' \
--description 'My Security Group' \
--vpc-id 'vpc-05627a462c11486de'
{
  "GroupId": "sg-0994929de0e8743f4",
  "SecurityGroupArn": "arn:aws:ec2:me-central-1:359416636630:security-group/sg-0994929de0e8743f4"
}
```

2. Inspect the security group (initially ingress will be empty or default):


```

@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-security-groups --group-ids sg-0994929de0e8743f4
{
  "SecurityGroups": [
    {
      "GroupId": "sg-0994929de0e8743f4",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-05627a462c11486de",
      "SecurityGroupArn": "arn:aws:ec2:me-central-1:359416636630:security-group/sg-0994929de0e8743f4",
      "OwnerId": "359416636630",
      "GroupName": "MySecurityGroup",
      "Description": "My Security Group",
      "IpPermissions": []
    }
  ]
}
...skipping...
{
  "SecurityGroups": [
    {
      "GroupId": "sg-0994929de0e8743f4",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-05627a462c11486de",
      "SecurityGroupArn": "arn:aws:ec2:me-central-1:359416636630:security-group/sg-0994929de0e8743f4",
      "OwnerId": "359416636630",
      "GroupName": "MySecurityGroup",
      "Description": "My Security Group",
      "IpPermissions": []
    }
  ]
}
}

```

3. Get your Codespace public IP (from inside the Codespace):

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $
4.240.39.192
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |

```

4. Authorize SSH inbound on port 22 from your Codespace IP:

```

@23-22411-057-sudo → /workspaces/exam_prep (main) $
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-0b2f73af225d92e1c",
      "GroupId": "sg-0994929de0e8743f4",
      "GroupOwnerId": "359416636630",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 22,
      "ToPort": 22,
      "CidrIpv4": "4.240.39.192/32",
      "SecurityGroupRuleArn": "arn:aws:ec2:me-central-1:359416636630:security-group-rule/sgr-0b2f73af225d92e1c"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $

```

5. Verify ingress rule appears:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $
{
  "SecurityGroups": [
    {
      "GroupId": "sg-0994929de0e8743f4",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-05627a462c11486de",
      "SecurityGroupArn": "arn:aws:ec2:me-central-1:359416636630:security-group/sg-0994929de0e8743f4",
      "OwnerId": "359416636630",
      "GroupName": "MySecurityGroup",
      "Description": "My Security Group",
      "IpPermissions": [
        {
          "IpProtocol": "tcp",
          "FromPort": 22,
          "ToPort": 22,
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "4.240.39.192/32"
            }
          ],
          "Ipv6Ranges": [],

```

6. Add an HTTP rule (port 80) using ip-permissions JSON:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-0e74eeae2a91bb3a7",
      "GroupId": "sg-0994929de0e8743f4",
      "GroupOwnerId": "359416636630",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "CidrIpv4": "4.240.39.192/32",
      "SecurityGroupRuleArn": "arn:aws:ec2:me-central-1:359416636630:security-group-rule/sgr-0e74eeae2a91bb3a7"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

7. Verify both ingress rules are present:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $
{
  "Return": true,
  "SecurityGroupRules": [
    {
      "SecurityGroupRuleId": "sgr-0e74eeae2a91bb3a7",
      "GroupId": "sg-0994929de0e8743f4",
      "GroupOwnerId": "359416636630",
      "IsEgress": false,
      "IpProtocol": "tcp",
      "FromPort": 80,
      "ToPort": 80,
      "CidrIpv4": "4.240.39.192/32",
      "SecurityGroupRuleArn": "arn:aws:ec2:me-central-1:359416636630:security-group-rule/sgr-0e74eeae2a91bb3a7"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
{
  "SecurityGroups": [
    {
      "GroupId": "sg-0994929de0e8743f4",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "PrefixListIds": []
    }
  ],
}
```

Task 4 — Create a key pair, describe key pairs, and launch EC2 instance

Goal: Create an ED25519 key pair, view it, and launch an EC2 instance using the key pair and the security group created earlier.

Steps:

1. Create the key pair and save the PEM file into the Codespace workspace:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 create-key-pair \
--key-name MyED25519Key \
--key-type ed25519 \
--key-format pem \
--query 'KeyMaterial' \
--output text > MyED25519Key.pem

An error occurred (InvalidKeyPair.Duplicate) when calling the CreateKeyPair operation: The keypair already exists
@23-22411-057-sudo → /workspaces/exam_prep (main) $ ls -l MyED25519Key.pem
-rw-rw-rw- 1 codespace codespace 0 Dec  5 11:50 MyED25519Key.pem
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

2. View created key pairs:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-key-pairs
{
  "KeyPairs": [
    {
      "KeyPairId": "key-0b172300531bccc37",
      "KeyType": "ed25519",
      "Tags": [],
      "CreateTime": "2025-12-05T11:32:25.952000+00:00",
      "KeyName": "MyED25519Key",
      "KeyFingerprint": "8Kj77B22v7uDNge+x6VRipIRhmnPE664nJfeRkcvdQ4="
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

3. Launch an EC2 instance (example values — replace IDs with ones from your account/region):

6. Stop, start and (optionally) terminate the instance:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 stop-instances --instance-ids i-0eac220f2c88b26bc
{
  "StoppingInstances": [
    {
      "InstanceId": "i-0eac220f2c88b26bc",
      "CurrentState": {
        "Code": 64,
        "Name": "stopping"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 start-instances --instance-ids i-0eac220f2c88b26bc
An error occurred (IncorrectInstanceState) when calling the StartInstances operation: The instance 'i-0eac220f2c88b26bc' is not in a state from which it can be started.
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-instances --instance-ids i-0eac220f2c88b26bc \
--query "Reservations[*].Instances[*].[InstanceId,State.Name,PublicIpAddress]" --output table
-----+-----+-----+-----+
| DescribeInstances |
+-----+-----+-----+-----+
| i-0eac220f2c88b26bc | stopped | None |
+-----+-----+-----+-----+
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 terminate-instances --instance-ids i-0eac220f2c88b26bc
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-0eac220f2c88b26bc",
      "CurrentState": {
        "Code": 48,
        "Name": "terminated"
      },
      "PreviousState": {
        "Code": 80,
        "Name": "stopped"
      }
    }
  ]
}
```

Task 5 — Understand AWS describe-* commands

Goal: Use describe commands to list and inspect AWS resources.

1.Run and understand these commands (run each, then capture screenshot immediately after):

aws ec2 describe-security-groups

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-security-groups
{
  "SecurityGroups": [
    {
      "GroupId": "sg-0d8508f3b485c64b8",
      "IpPermissionsEgress": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [],
          "IpRanges": [
            {
              "CidrIp": "0.0.0.0/0"
            }
          ],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        }
      ],
      "VpcId": "vpc-05627a462c11486de",
      "SecurityGroupArn": "arn:aws:ec2:me-central-1:359416636630:security-group/sg-0d8508f3b485c64b8",
      "OwnerId": "359416636630",
      "GroupName": "default",
      "Description": "default VPC security group",
      "IpPermissions": [
        {
          "IpProtocol": "-1",
          "UserIdGroupPairs": [
            {
              "UserId": "359416636630",
              "GroupId": "sg-0d8508f3b485c64b8"
            }
          ],
          "IpRanges": [],
          "Ipv6Ranges": [],
          "PrefixListIds": []
        },
        {
          "IpProtocol": "tcp",
          "FromPort": 22,
          "ToPort": 22,
          "UserIdGroupPairs": [],

```

aws ec2 describe-vpcs

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-vpcs
{
  "Vpcs": [
    {
      "OwnerId": "359416636630",
      "InstanceTenancy": "default",
      "CidrBlockAssociationSet": [
        {
          "AssociationId": "vpc-cidr-assoc-0185d7fdfa697e471",
          "CidrBlock": "172.31.0.0/16",
          "CidrBlockState": {
            "State": "associated"
          }
        }
      ],
      "IsDefault": true,
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "VpcId": "vpc-05627a462c11486de",
      "State": "available",
      "CidrBlock": "172.31.0.0/16",
      "DhcpOptionsId": "dopt-07a746a94dc2af666"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

aws ec2 describe-subnets

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-subnets
{
  "Subnets": [
    {
      "AvailabilityZoneId": "mec1-az3",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "359416636630",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-05854ba6444478aac",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,
        "EnableResourceNameDnsAAAARecord": false
      },
      "BlockPublicAccessStates": {
        "InternetGatewayBlockMode": "off"
      },
      "SubnetId": "subnet-05854ba6444478aac",
      "State": "available",
      "VpcId": "vpc-05627a462c11486de",
      "CidrBlock": "172.31.0.0/20",
      "AvailableIpAddressCount": 4090,
      "AvailabilityZone": "me-central-1c",
      "DefaultForAz": true,
      "MapPublicIpOnLaunch": true
    },
    {
      "AvailabilityZoneId": "mec1-az1",
      "MapCustomerOwnedIpOnLaunch": false,
      "OwnerId": "359416636630",
      "AssignIpv6AddressOnCreation": false,
      "Ipv6CidrBlockAssociationSet": [],
      "SubnetArn": "arn:aws:ec2:me-central-1:359416636630:subnet/subnet-0ae203a686bd96349",
      "EnableDns64": false,
      "Ipv6Native": false,
      "PrivateDnsNameOptionsOnLaunch": {
        "HostnameType": "ip-name",
        "EnableResourceNameDnsARecord": false,

```

aws ec2 describe-instances

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-instances
{
  "Reservations": [
    {
      "ReservationId": "r-05f050d79728288ab",
      "OwnerId": "359416636630",
      "Groups": [],
      "Instances": [
        {
          "Architecture": "x86_64",
          "BlockDeviceMappings": [],
          "ClientToken": "91a00bad-b131-4dca-8022-2467874c80d2",
          "EbsOptimized": false,
          "EnaSupport": true,
          "Hypervisor": "xen",
          "NetworkInterfaces": [],
          "RootDeviceName": "/dev/xvda",
          "RootDeviceType": "ebs",
          "SecurityGroups": [],
          "StateReason": {
            "Code": "Client.UserInitiatedShutdown",
            "Message": "Client.UserInitiatedShutdown: User initiated shutdown"
          },
          "Tags": [
            {
              "Key": "Name",
              "Value": "MyServer"
            }
          ],
          "VirtualizationType": "hvm",
          "CpuOptions": {
            "CoreCount": 1,
            "ThreadsPerCore": 2
          },
          "CapacityReservationSpecification": {
            "CapacityReservationPreference": "open"
          },
          "HibernationOptions": {
            "Configured": false
          },
          "MetadataOptions": {
```

aws ec2 describe-regions

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-regions
```

```
{
  "Regions": [
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "ap-south-1",
      "Endpoint": "ec2.ap-south-1.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "eu-north-1",
      "Endpoint": "ec2.eu-north-1.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "eu-west-3",
      "Endpoint": "ec2.eu-west-3.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "eu-west-2",
      "Endpoint": "ec2.eu-west-2.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "eu-west-1",
      "Endpoint": "ec2.eu-west-1.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "ap-northeast-3",
      "Endpoint": "ec2.ap-northeast-3.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "ap-northeast-2",
      "Endpoint": "ec2.ap-northeast-2.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "ap-northeast-1",
      "Endpoint": "ec2.ap-northeast-1.amazonaws.com"
    }
  ],
  "IsTruncated": false
}
```

```
:...skipping...
```

```
{
  "Regions": [
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "ap-south-1",
      "Endpoint": "ec2.ap-south-1.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "eu-north-1",
      "Endpoint": "ec2.eu-north-1.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "eu-west-3",
      "Endpoint": "ec2.eu-west-3.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "eu-west-2",
      "Endpoint": "ec2.eu-west-2.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "eu-west-1",
      "Endpoint": "ec2.eu-west-1.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "ap-northeast-3",
      "Endpoint": "ec2.ap-northeast-3.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "ap-northeast-2",
      "Endpoint": "ec2.ap-northeast-2.amazonaws.com"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "RegionName": "ap-northeast-1",
      "Endpoint": "ec2.ap-northeast-1.amazonaws.com"
    }
  ],
  "IsTruncated": false
}
```

aws ec2 describe-availability-zones


```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-availability-zones
{
  "AvailabilityZones": [
    {
      "OptInStatus": "opt-in-not-required",
      "Messages": [],
      "RegionName": "me-central-1",
      "ZoneName": "me-central-1a",
      "ZoneId": "mec1-az1",
      "GroupName": "me-central-1-zg-1",
      "NetworkBorderGroup": "me-central-1",
      "ZoneType": "availability-zone",
      "GroupLongName": "Middle East (UAE) 1",
      "State": "available"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "Messages": [],
      "RegionName": "me-central-1",
      "ZoneName": "me-central-1b",
      "ZoneId": "mec1-az2",
      "GroupName": "me-central-1-zg-1",
      "NetworkBorderGroup": "me-central-1",
      "ZoneType": "availability-zone",
      "GroupLongName": "Middle East (UAE) 1",
      "State": "available"
    },
    {
      "OptInStatus": "opt-in-not-required",
      "Messages": [],
      "RegionName": "me-central-1",
      "ZoneName": "me-central-1c",
      "ZoneId": "mec1-az3",
      "GroupName": "me-central-1-zg-1",
      "NetworkBorderGroup": "me-central-1",
      "ZoneType": "availability-zone",
      "GroupLongName": "Middle East (UAE) 1",
      "State": "available"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

Task 6 — IAM: create group, user, attach policies, create console login & keys

1. Create group:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam create-group --group-name MyGroupCli
{
  "Group": {
    "Path": "/",
    "GroupName": "MyGroupCli",
    "GroupId": "AGPAVHLXGUTLDN7IIXOWX",
    "Arn": "arn:aws:iam::359416636630:group/MyGroupCli",
    "CreateDate": "2025-12-05T13:32:49+00:00"
  }
}
```

2. Get group details:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam get-group --group-name MyGroupCli
{
  "Users": [],
  "Group": {
    "Path": "/",
    "GroupName": "MyGroupCli",
    "GroupId": "AGPAVHLXGUTLDN7IIXOWX",
    "Arn": "arn:aws:iam::359416636630:group/MyGroupCli",
    "CreateDate": "2025-12-05T13:32:49+00:00"
  }
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

3. Create user:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam create-user --user-name MyUserCli
{
  "User": {
    "Path": "/",
    "UserName": "MyUserCli",
    "UserId": "AIDAVHLXGUTLHJZPY2JYG",
    "Arn": "arn:aws:iam::359416636630:user/MyUserCli",
    "CreateDate": "2025-12-05T13:33:50+00:00"
  }
}
```

4. Get user details:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam get-user --user-name MyUserCli
{
  "User": {
    "Path": "/",
    "UserName": "MyUserCli",
    "UserId": "AIDAVHLXGUTLHJZPY2JYG",
    "Arn": "arn:aws:iam::359416636630:user/MyUserCli",
    "CreateDate": "2025-12-05T13:33:50+00:00"
  }
}
```

5. Add user to group:

6. See group again:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam add-user-to-group --user-name MyUserCli --group-name MyGroupCli
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam get-group --group-name MyGroupCli
{
  "Users": [
    {
      "Path": "/",
      "UserName": "MyUserCli",
      "UserId": "AIDAVHLXGUTLHJZPY2JYG",
      "Arn": "arn:aws:iam::359416636630:user/MyUserCli",
      "CreateDate": "2025-12-05T13:33:50+00:00"
    }
  ],
  "Group": {
    "Path": "/",
    "GroupName": "MyGroupCli",
    "GroupId": "AGPAVHLXGUTLDN7IIX0WX",
    "Arn": "arn:aws:iam::359416636630:group/MyGroupCli",
    "CreateDate": "2025-12-05T13:32:49+00:00"
  }
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

7. List policies that mention EC2:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam list-policies \
--query "Policies[?contains(PolicyName, 'EC2')].{Name:PolicyName}" \
--output text
AmazonEC2FullAccess
AmazonEC2ReadOnlyAccess
AmazonElasticMapReduceforEC2Role
AmazonEC2RoleforDataPipelineRole
AmazonEC2ContainerServiceforEC2Role
AmazonEC2ContainerServiceRole
AmazonEC2RoleforAWSCodeDeploy
AmazonEC2RoleforSSM
CloudWatchActionsEC2Access
AmazonEC2ContainerRegistryReadOnly
AmazonEC2ContainerRegistryPowerUser
AmazonEC2ContainerRegistryFullAccess
AmazonEC2ContainerServiceAutoscaleRole
AmazonEC2SpotFleetAutoscaleRole
AWSElasticBeanstalkCustomPlatformforEC2Role
AmazonEC2ContainerServiceEventsRole
AmazonEC2SpotFleetTaggingRole
AWSEC2SpotServiceRolePolicy
AWSServiceRoleForEC2ScheduledInstances
AWSEC2SpotFleetServiceRolePolicy
AWSApplicationAutoscalingEC2SpotFleetRequestPolicy
AWSEC2FleetServiceRolePolicy
AWSAutoScalingPlansEC2AutoScalingPolicy
EC2InstanceConnect
AmazonEC2RolePolicyForLaunchWizard
EC2InstanceProfileForImageBuilder
EC2FleetTimeShiftableServiceRolePolicy
AmazonEC2RoleforAWSCodeDeployLimited
EC2InstanceProfileForImageBuilderECRContainerBuilds
AWSApplicationMigrationEC2Access
AWSEC2CapacityReservationFleetRolePolicy
EC2FastLaunchServiceRolePolicy
AmazonSSMManagedEC2InstanceDefaultPolicy
AWSFaultInjectionSimulatorEC2Access
EC2ImageBuilderLifecycleExecutionPolicy
AWSEC2VssSnapshotPolicy
EC2FastLaunchFullAccess
AmazonEC2ContainerRegistryPullOnly
DeclarativePoliciesEC2Report
AmazonEC2ImageReferencesAccessPolicy
AWSEC2CapacityManagerServiceRolePolicy
AWSEC2SqlHaServiceRolePolicy
AWSEC2SqlHaInstancePolicy
AWSLambdaManagedEC2ResourceOperator
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

8. Get ARN for AmazonEC2FullAccess (example query):

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam list-policies --query 'Policies[?PolicyName=='AmazonEC2FullAccess'].{Name:PolicyName, ARN:Arn}' --output table
```

ListPolicies	
ARN	Name
arn:aws:iam::aws:policy/AmazonEC2FullAccess	AmazonEC2FullAccess

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

9. Attach policy to group (use the ARN you retrieved):

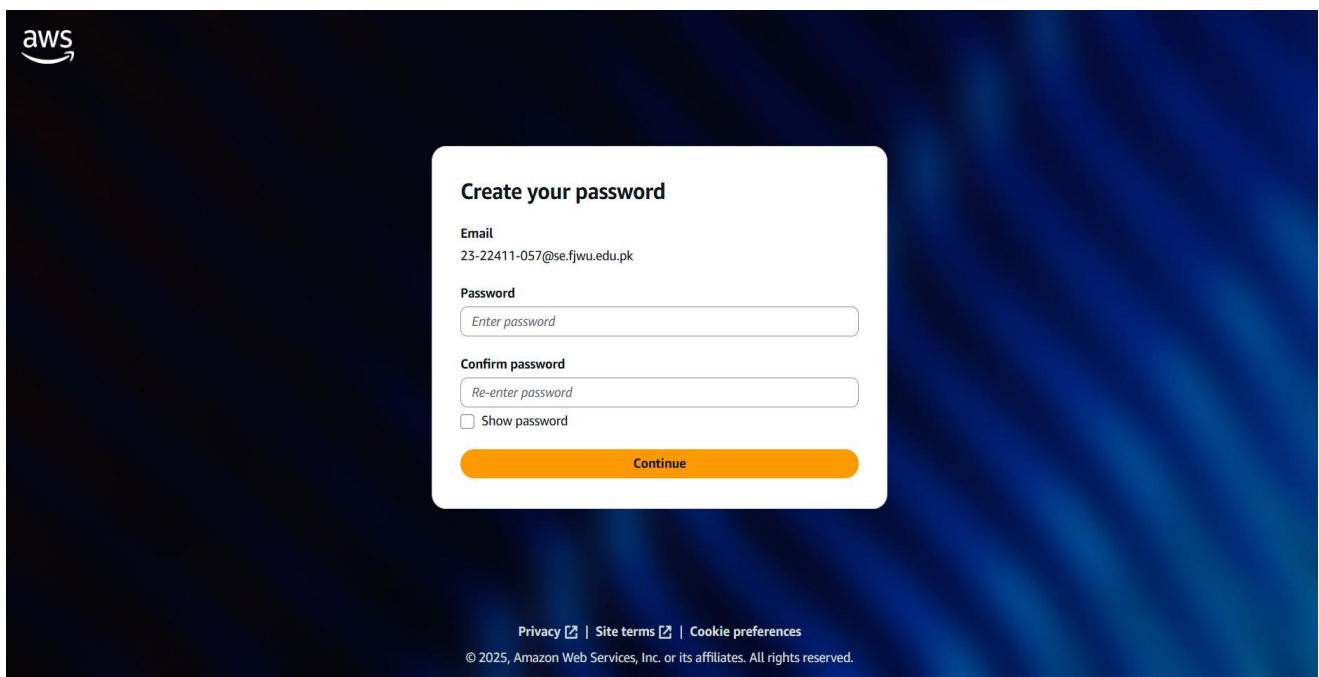
10. List attached policies for the group:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam attach-group-policy \
--group-name MyGroupCli \
--policy-arn arn:aws:iam::aws:policy/AmazonEC2FullAccess
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam list-attached-group-policies --group-name MyGroupCli
{
  "AttachedPolicies": [
    {
      "PolicyName": "AmazonEC2FullAccess",
      "PolicyArn": "arn:aws:iam::aws:policy/AmazonEC2FullAccess"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

11. Create a console login profile for the user:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam create-login-profile \
--user-name MyUserCli \
--password "MyStrongP@ssw0rd123" \
--password-reset-required
{
  "LoginProfile": {
    "UserName": "MyUserCli",
    "CreateDate": "2025-12-05T13:42:48+00:00",
    "PasswordResetRequired": true
  }
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

12. If the user cannot change password, attach IAMUserChangePassword to the group temporarily:



```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam attach-group-policy --group-name MyGroupCli --policy-arn arn:aws:iam::aws:policy/IAMUserChangePassword
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam detach-group-policy --group-name MyGroupCli --policy-arn arn:aws:iam::aws:policy/IAMUserChangePassword
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam attach-group-policy \
--group-name MyGroupCli \
--policy-arn arn:aws:iam::aws:policy/IAMUserChangePassword
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

13. Create access keys for the user (save AccessKeyId and SecretAccessKey securely):


```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam create-access-key --user-name MyUserCli
{
  "AccessKey": {
    "UserName": "MyUserCli",
    "AccessKeyId": "AKIAVHLXGUTLIYJR0TR7",
    "Status": "Active",
    "SecretAccessKey": "WC5kf/boKuYZ9LW46F/39ew0vf5Jj876TGKS3Mcq",
    "CreateDate": "2025-12-05T13:53:03+00:00"
  }
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

14. List access keys:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam list-access-keys --user-name MyUserCli
{
  "AccessKeyMetadata": [
    {
      "UserName": "MyUserCli",
      "AccessKeyId": "AKIAVHLXGUTLIYJR0TR7",
      "Status": "Active",
      "CreateDate": "2025-12-05T13:53:03+00:00"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

15. (Don't) Delete access key:

16. Use environment variables to authenticate as that user in the Codespace:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam delete-access-key --user-name MyUserCli --access-key-id AKIAVHLXGUTLIYJR0TR7
@23-22411-057-sudo → /workspaces/exam_prep (main) $ export AWS_ACCESS_KEY_ID=AKIAVHLXGUTLDQ5QTAH3
@23-22411-057-sudo → /workspaces/exam_prep (main) $ export AWS_SECRET_ACCESS_KEY=SfK/OARpFV2aY5v/AbzwpMv1sXhP05uHSYs+0+TT
@23-22411-057-sudo → /workspaces/exam_prep (main) $ printenv | grep AWS_
AWS_SECRET_ACCESS_KEY=SfK/OARpFV2aY5v/AbzwpMv1sXhP05uHSYs+0+TT
AWS_ACCESS_KEY_ID=AKIAVHLXGUTLDQ5QTAH3
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam get-user --user-name MyUserCli
{
  "User": {
    "Path": "/",
    "UserName": "MyUserCli",
    "UserId": "AIDAVHLXGUTLHJZPY2JYG",
    "Arn": "arn:aws:iam::359416636630:user/MyUserCli",
    "CreateDate": "2025-12-05T13:33:50+00:00"
  }
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

Task 8 — Use --query to format outputs for reporting

Goal: Extract useful fields in table format for reporting.

Examples (run each and take a screenshot immediately after):

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-instances \
  --filters "Name=tag:Name,Values=MyServer" \
  --query "Reservations[*].Instances[*].[InstanceId,PublicIpAddress,Tags[?Key=='Name'].Value|[0]]" \
  --output table
```

DescribeInstances		
i-0f66592f6efd44a5d	51.112.195.228	MyServer

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-instances \
  --query "Reservations[*].Instances[*].[InstanceId,State.Name]" \
  --output table
```

DescribeInstances	
i-0f66592f6efd44a5d	running

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-instances \
--query "Reservations[*].Instances[*].[InstanceId,InstanceType,Placement.AvailabilityZone]" \
--output table
```

DescribeInstances		
i-0f66592f6efd44a5d	t3.micro	me-central-1c

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $
```

Cleanup — Remove resources to avoid charges

1. Terminate instances:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 terminate-instances --instance-ids i-0f66592f6efd44a5d
```

```
{
  "TerminatingInstances": [
    {
      "InstanceId": "i-0f66592f6efd44a5d",
      "CurrentState": {
        "Code": 32,
        "Name": "shutting-down"
      },
      "PreviousState": {
        "Code": 16,
        "Name": "running"
      }
    }
  ]
}
```

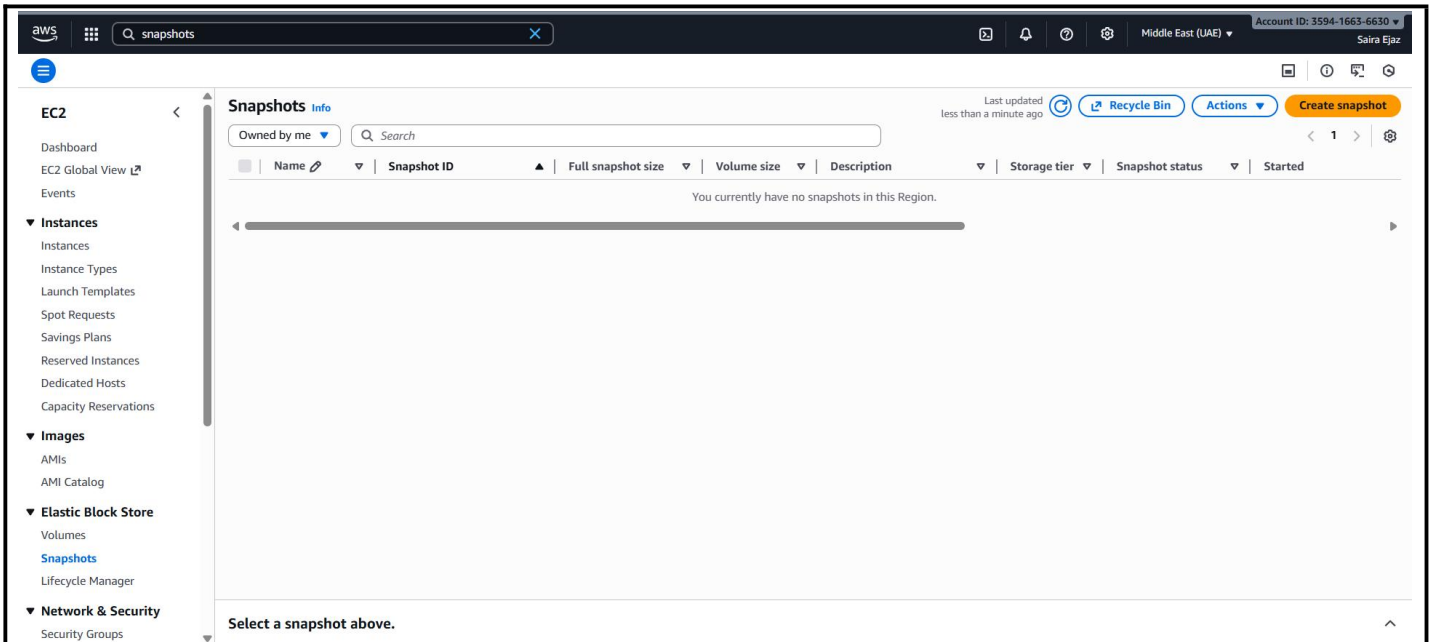
```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 describe-instances --query "Reservations[*].Instances[*].[InstanceId,State.Name]" --output table
```

DescribeInstances	
i-0f66592f6efd44a5d	terminated

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ |
```

2. Delete EBS Volumes (if any)

The screenshot displays the AWS Management Console interface for the EC2 Volumes section. The top navigation bar includes the AWS logo, a search bar, and account details (Account ID: 3594-1663-6630, Middle East (UAE)). The left sidebar lists various EC2 services: Dashboard, EC2 Global View, Events, Instances (with sub-links for Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, and Capacity Reservations), Images (with sub-links for AMIs and AMI Catalog), Elastic Block Store (with sub-links for Volumes, Snapshots, and Lifecycle Manager), and Network & Security (with sub-link for Security Groups). The main content area is titled 'Volumes' and shows a message: 'You currently have no volumes in this region'. Below this, there is a 'Snapshot summary' section indicating '0 / 0' recently backed up volumes out of a total of 0 volumes. A 'Data Lifecycle Manager default policy for EBS Snapshots status' section notes that no default policy is set up, with a link to 'Create policy'. The bottom of the console shows a footer with copyright information (© 2025, Amazon Web Services, Inc. or its affiliates) and links for Privacy, Terms, and Cookie preferences.



3. Delete security group and key pair:

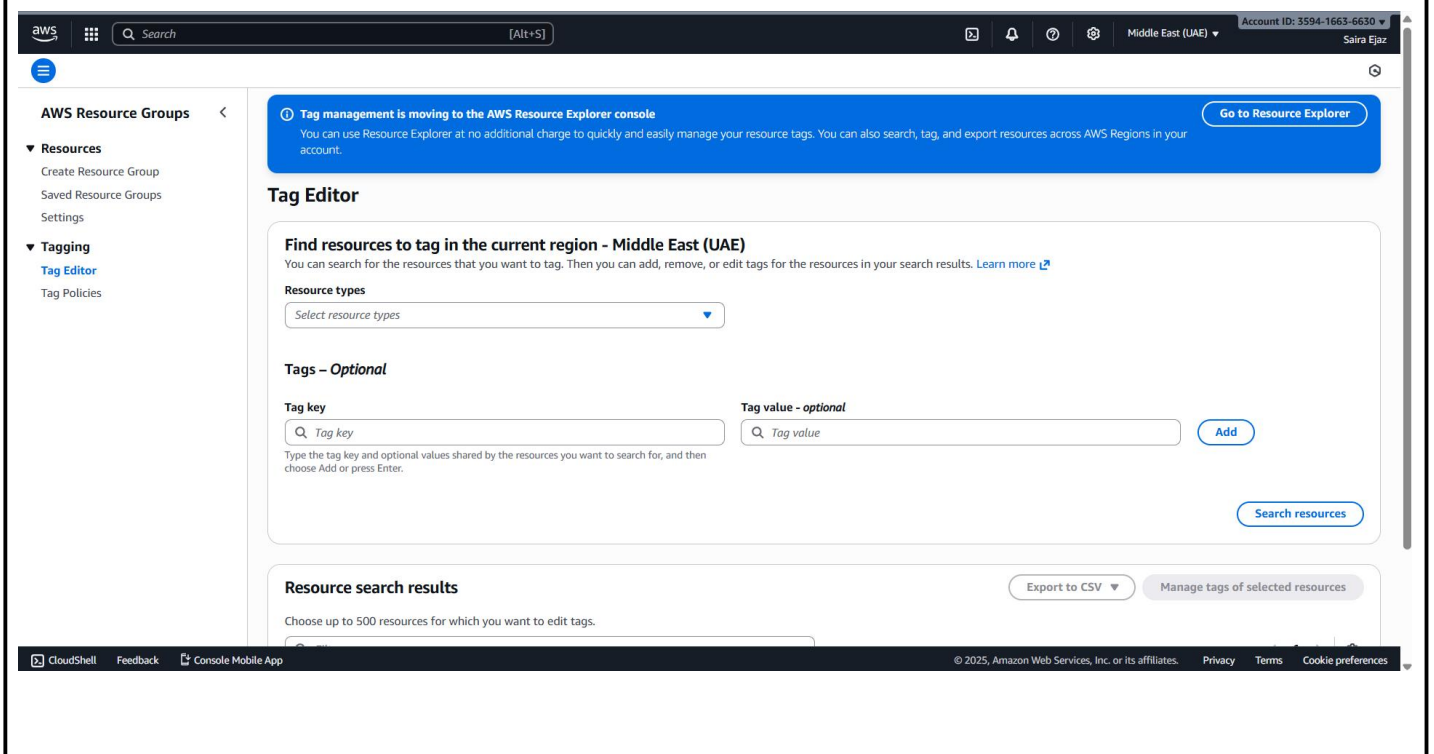
```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws ec2 delete-key-pair --key-name MyED25519Key
{
  "Return": true,
  "KeyPairId": "key-0207823d468fe6529"
}
```

4. Remove IAM users, access keys, groups:

```
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam delete-login-profile --user-name MyUserCli
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam remove-user-from-group --user-name MyUserCli --group-name MyGroupCli
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam delete-user --user-name MyUserCli
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam detach-group-policy --group-name MyGroupCli --policy-arn arn:aws:iam::aws:policy/AmazonEC2FullAccess
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam delete-group --group-name MyGroupCli

An error occurred (DeleteConflict) when calling the DeleteGroup operation: Cannot delete entity, must detach all policies first.
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam list-attached-group-policies --group-name MyGroupCli
{
  "AttachedPolicies": [
    {
      "PolicyName": "IAMUserChangePassword",
      "PolicyArn": "arn:aws:iam::aws:policy/IAMUserChangePassword"
    }
  ]
}
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam detach-group-policy --group-name MyGroupCli --policy-arn arn:aws:iam::aws:policy/IAMUserChangePassword
@23-22411-057-sudo → /workspaces/exam_prep (main) $ aws iam delete-group --group-name MyGroupCli
```

5. Clean Summary



aws

Q Search

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🔔

⚙️

🌐 Global

Account ID: 3594-1663-6630

Saiza Ejaz

Billing and Cost Management

Bills

Billing and Cost Management

Billing View

Home

Getting Started

Dashboards

Billing and Payments

Bills

Payments

Credits

Purchase Orders

Cost and Usage Analysis

Cost Explorer

Cost Explorer Saved Reports

Cost Anomaly Detection

Free Tier

Data Exports

Customer Carbon Footprint Tool

Cost Organization

Cost Categories

Cost Allocation Tags

Billing Conductor

Budgets and Planning

Charges by service

Charges by account

Invoices

Savings

Taxes by service

Amazon Web Services, Inc. charges by service

Expand all

Total active services10

Total pre-tax service charges in USD0.00

Filter by service name or region name

Description	Usage Quantity	Amount in USD
Data Transfer		USD 0.00
Elastic Compute Cloud		USD 0.00
Glue		USD 0.00
Key Management Service		USD 0.00
Location Service		USD 0.00
Secrets Manager		USD 0.00
Simple Notification Service		USD 0.00
Simple Queue Service		USD 0.00
Simple Storage Service		USD 0.00
Virtual Private Cloud		USD 0.00
Total tax		USD 0.00

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